



OWASP Juice Shop + DVWA Lab Setup

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Goal

Build and compare vulnerable web apps (Juice Shop & DVWA) across isolated VLANs for educational purposes in a controlled lab using Windows Server 2022, Ubuntu, and Cisco RV340.



Environment Overview

Component	Details
Windows Server 2022	Hosting DVWA (XAMPP) + OWASP Juice Shop
Ubuntu VM	Jump box for accessing apps and performing attacks
Cisco RV340	Router providing VLAN segmentation for isolation
VLAN 10	172.20.10.0/24 network for test environment
VLAN 999	10.10.99.0/24 alternate test network
GitHub Repo	Contains cyber-dash scripts and upload portal



Actions Completed



DVWA Setup on Windows Server (XAMPP)

- Installed [XAMPP](#)
 - Extracted DVWA-master.zip to C:\xampp\htdocs\DVWA-master
 - Configured config.inc.php
 - Started Apache & MySQL
 - Resolved setup issues:
 - File/folder permissions
 - PHP allow_url_fopen, allow_url_include
 - Accessed DVWA at:
 - http://172.20.10.200/DVWA-master/login.php
 - Confirmed command injection functionality using:
 - 127.0.0.1 && whoami
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Network Segmentation

- Configured Cisco RV340 VLANs:
 - VLAN 10: 172.20.10.0/24
 - VLAN 999: 10.10.99.0/24
 - Updated VM NICs to be **tagged on VLAN 10**
 - Set Ubuntu VM as a **jump box**
 - Added default route from Ubuntu to 10.10.10.1 to allow cross-VLAN access
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Python Upload Server (cyber-dash)

- Located at: /opt/cyber-dash/scripts/server.py
 - Modified Python HTTP server to bind to 172.20.10.x:
 - `server_address = ("172.20.10.112", 8081)`
 - Used for transferring cheat files and uploads from Windows → Linux
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OWASP Juice Shop Setup (Windows)

- Installed:
 - Git for Windows
 - Python (for future HTTP server use)
 - Node.js & npm (required for Juice Shop)
 - Cloned Juice Shop repo:
 - `git clone https://github.com/juice-shop/juice-shop.git`
 - `cd juice-shop`
 - `npm install`
 - `npm start`
 - Opened port 3000 through Windows Defender Firewall:
 - `netsh advfirewall firewall add rule name="Juice Shop 3000" dir=in action=allow protocol=TCP localport=3000`
 - Accessed at:
 - `http://172.20.10.200:3000`
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Next Steps

- Document and compare output differences in DVWA between Ubuntu & Windows
- Launch Juice Shop CTF mode
- Automate setup using batch or shell scripts
- Begin recording video content for educational series